Supporting Information

Exploring the Interactions between Methyl Methacrylate Polymers and the Bacterial Outer Membrane via Coarse-Grained Molecular Dynamics Simulations

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Table S1. Chemical and physical properties of the studied methyl-methacrylate polymers.

Polymer	Properties				
	Ionic character[a]	Charge[b]	Ionizable group[a]	Hydrophilicity[c,d,e]	Volume/nm^3 [f]
pDMAEMA	cationic	+96	tertiary amine	hydrophilic	1356.6
pMETAC	cationic	+96	quaternary ammonium	highly hydrophilic	1487.6
pMEDSAH	zwitterionic	0	quaternary ammonium+ sulfonate	highly hydrophilic	1315.1
pSPMA	anionic	-96	sulfonate	hydrophilic	1218.8

[a]Ref.1. [b]The charge of these polyelectrolyte polymers corresponds to the sum of the individual charges of their constituent monomers. [c]Ref.2. [d]Ref.3. [e]Ref.4. [f]The volume of the polymers was estimated as the excluded volume of the solvent from the solvent assessment surface area (sasa) calculation implemented in GROMACS version 2019.4 (ref.5,6).

Table S2. Composition of the coarse-grained bacterial outer membrane (OM) model of *Escherichia coli*.

	Number of species[a]	
Components	**Outer leaflet**	**Inner leaflet**
rough LPS[b]	560	0
DPPE[c]	0	1260
DPPG[d]	0	420
Ions		
Ca^{2+} [e]	3012	
Cl^{-} [f]	4	
Solvent		
TW[g]	616312	

[a]All species were represented with the MARTINI 3 force field (FF) (ref.7). [b]Rough lipopolysaccharide parameterized as discussed in ref. 8. [c]1,2-dipalmitoyl-sn-glycero-3-phosphoethanolamine. [d]1,2-dipalmitoyl-sn-glycero-3-phospho-rac-glycerol. [e]The Ca^{2+} ion was modeled by the divalent bead CA (ref.

7). [f]The Cl^- ion was modeled by the monovalent bead (ref. 7). [g]The water molecules were modeled with the tiny bead water (ref. 7).

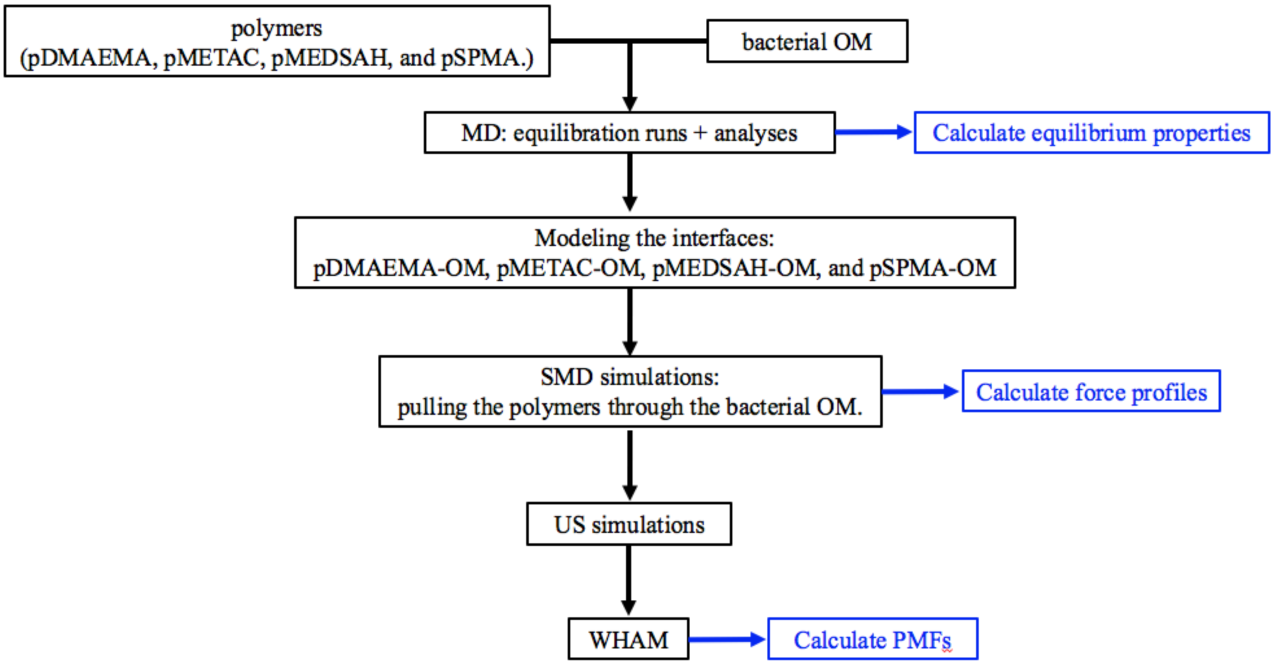

Figure S1 Computational workflow conducted in this work using the following methods implemented in the GROMACS software version 2019.4:[8] steered molecular dynamics (SMD),[9] umbrella sampling (US) method,[10] and Weighted Histogram Analysis Method (WHAM).[11]

Before analyzing the interaction between the free polymers and the bacterial OM, we firstly evaluated the equilibration of the isolated systems in aqueous solution at 310 K and 1 bar. Regarding the free polymers, the convergence in the time evolution of the root mean square deviation (RMSD) and the radius of gyration (R_g) in Figure S2 suggests structural stability of these systems during the 1.0 μs simulations.

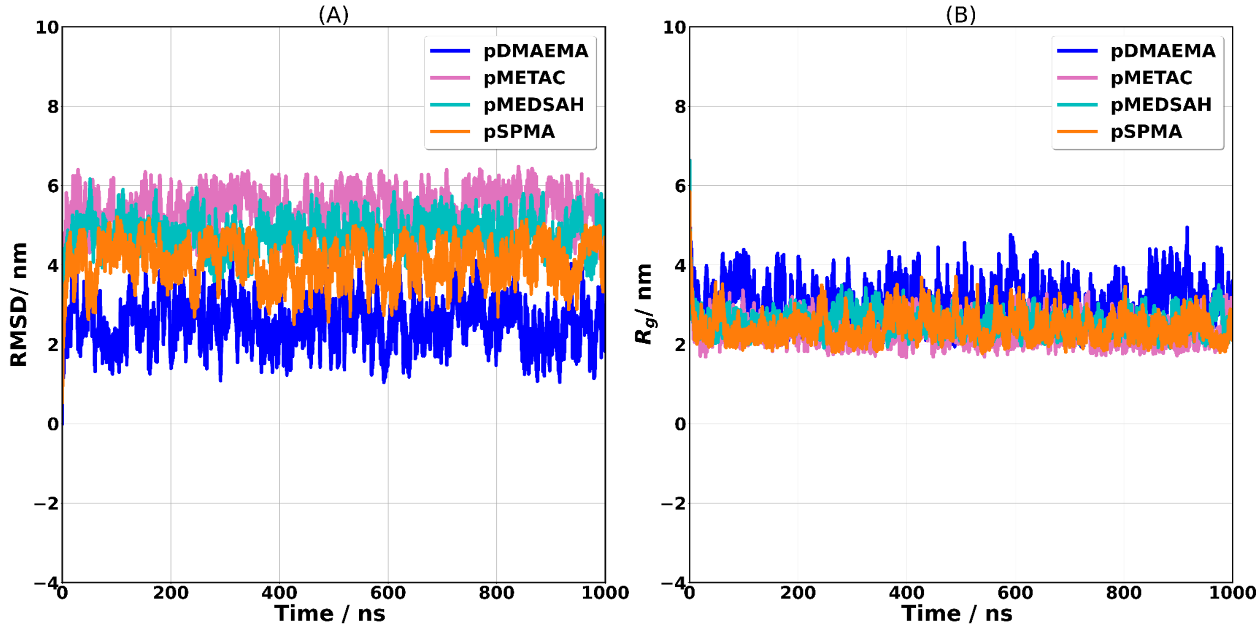

Figure S2 Time evolution of the root mean square deviation (RMSD) and the radius of gyration (R_g) of the free polymers pDMAEMA, pMETAC, pMEDSAH, and pSPMA in aqueous solution during the 1.0 μs simulations: RMSD (A) and R_g (B).

Data from Table S3 reinforce this equilibration since the standard deviation values of these measures represent about 13 % of the average RMSD and 13.3 % of the average R_g for all polymers. In particular, the R_g of the polymers pMETAC, pMEDSAH, and pSPMA indicates that the structures of these systems are more compact than the one of pDMAEMA with a difference of 0.63 nm in the R_g between these species. This behavior may be related to the higher molecular volume of pMETAC, pMEDSAH, and pSPMA and their charges that may have increased the number of solute-solute interactions favoring the level of compaction of these polymeric chains.

Table S3. Structural properties of the free polymers pDMAEMA, pMETAC, pMEDSAH, and pSPMA in aqueous solution calculated as average values from molecular dynamics simulations of 1.0 μs.

Polymer	RMSD / nm [a]	R_g / nm [b]
pDMAEMA	2.56±0.58	3.04±0.48
pMETAC	5.41±0.47	2.28±0.30
pMEDSAH	4.77±0.41	2.54±0.27
pSPMA	4.08±0.51	2.41±0.33

[a]Root mean square deviation. [b]Radius of gyration.

For the bacterial OM model, the stable behavior of the temporal variation of both area per lipopolysaccharide (A_{LPS}) and per DPPE and DPPG ($A_{DPPE/DPPG}$) in Figure S3A and Figure S3B suggests the convergence of these properties and, therefore, the structural stability of this asymmetric bilayer after 1.0 μs. The average A_{LPS} per acyl chain (0.298 nm^2) is in agreement with the range of theoretical and experimental values reported per acyl tail (0.26-0.32 nm^2).[8,12-17] Finally, in addition to the stable time series of the membrane thickness in Figure S3C, the average mass density profile in Figure S3D also reflects the structural stabilization of this bacterial membrane prototype. In particular, although the solvent was also represented at the CG resolution, Figure S3D shows that the saccharides core of the LPSs and the polar heads of the phospholipids (DPPE and DPPG) are fully hydrated, which indicates that the tiny water beads (TW) of the MARTINI 3 force field[7] were able to properly represent the solvation of these molecules in agreement with the atomistic models.[14-16]

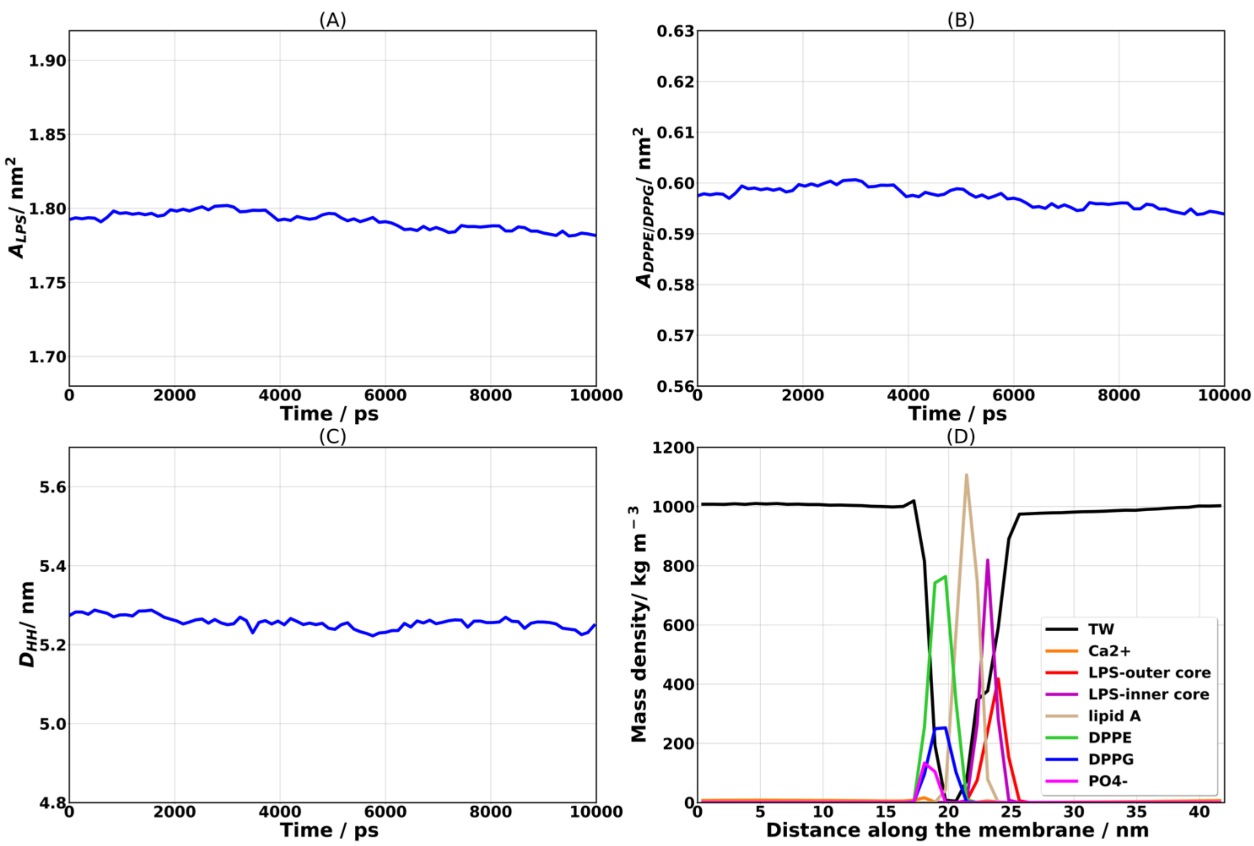

Figure S3 Temporal variation of the structural properties of the bacterial outer membrane (OM) model during the last 10 ns of the 6.0 μs molecular dynamics simulation (equilibration run): area per lipid of the LPS molecules (A), area per lipid of the phospholipids DPPE and DPPG (B), and membrane thickness (C). Average mass density profile (D).

Table S4. Average values of the structural properties of the bacterial outer membrane (OM) model calculated from the last 10 ns of the 6.0 μs molecular dynamics simulation (equilibration run) at 310 K and 1.0 bar.

Property / unity	Average ± Std.
A_{LPS} / nm^2 [a]	1.792 ± 0.006 (1.30-1.92)[d]
$A_{DPPE/DPPG} / nm^2$ [b]	0.597± 0.002 (0.605[e], 0.588[f], 0.55[g])
D_{HH} / nm [c]	5.257 ± 0.015

[a]Area per LPS molecule. [b]Area per lipid (DPPE and DPPG). [c]Membrane thickness is calculated as the average distance between the center of mass referring to the phosphates groups in the phospholipids and the center of mass referring to the outer core of the LPS molecules. [d]Ref. 7,12-17. [e]Ref. 18, value at 342 K for a homogeneous DPPE membrane.[f]Ref. 19, value at 333 K for a homogeneous DPPE membrane. [g]Ref. 20, value at 342 K.

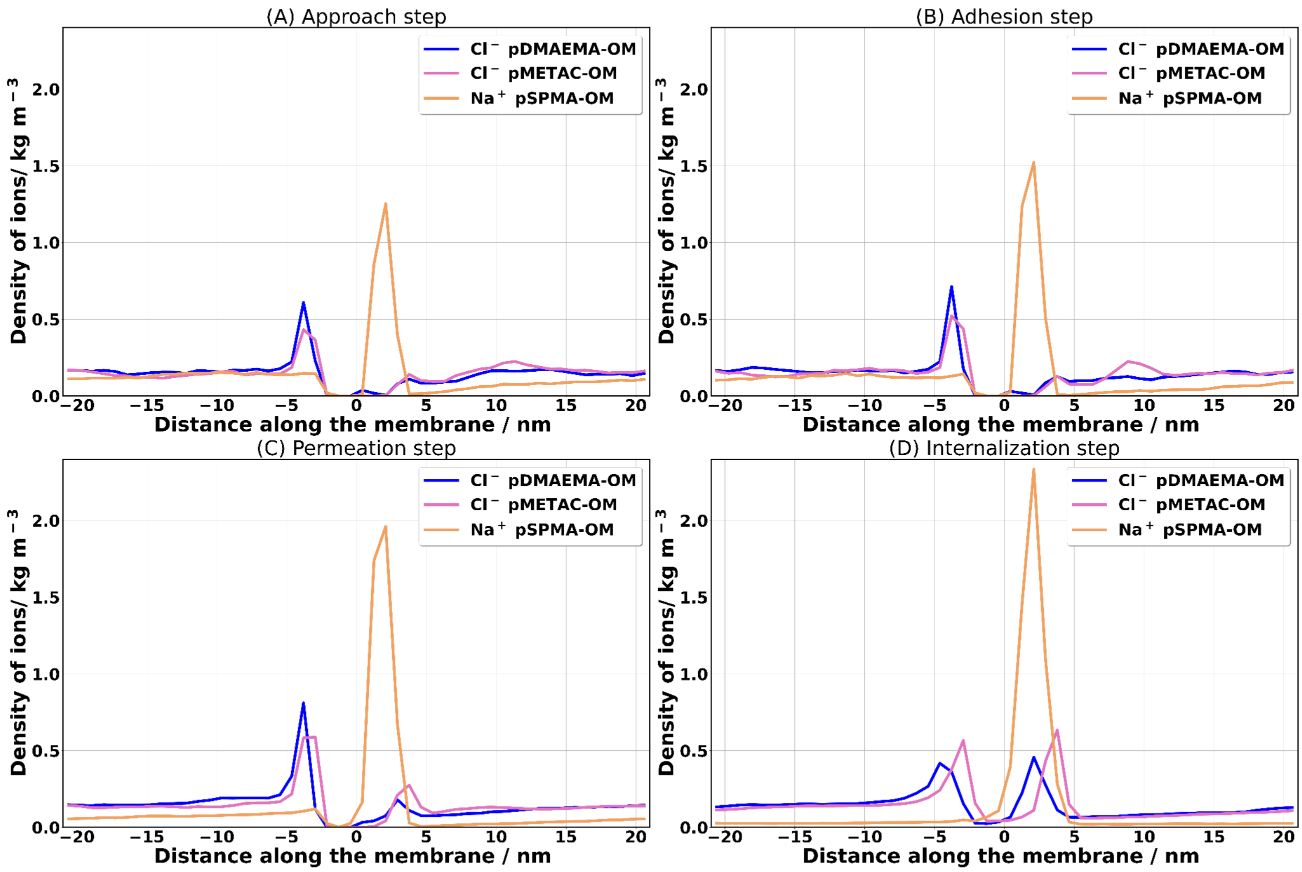

Figure S4 Density profiles of the Cl^- and Na^+ ions during the translocation process of the polymers (pDMAEMA, pMETAC, pMEDSAH, and pSPMA) through the bacterial outer membrane (OM): approach step (A), adhesion step (B), permeation step (C), and internalization step (D). Each profile was calculated as an average from three independent runs.

Table S5. Coordination number (CN) of water beads (TW bead of the MARTINI 3 force field) at a 4 nm radial distance of the polymers (pDMAEMA, pMETAC, pMEDSAH, and pSPMA) during their translocation through the bacterial outer membrane (OM).

System	CN(TW)*			
	Approach	**Adhesion**	**Permeation**	**Internalization**
pDMAEMA-OM	4295 ± 91	4016 ± 296	2993 ± 99	2987 ± 434
pMETAC-OM	4266 ± 15	4248 ± 8	3320 ± 89	3328 ± 52
pMEDSAH-OM	4147 ± 12	3852 ± 39	2978 ± 47	3747 ± 31
pSPMA-OM	4340 ± 19	4303 ± 7	3480 ± 24	3783 ± 15

*The CN was calculated from the radial distribution functions (g(r)) defined between the center of mass of the free polymers and the water beads (TW). Each value represents the average of three independent replicates.

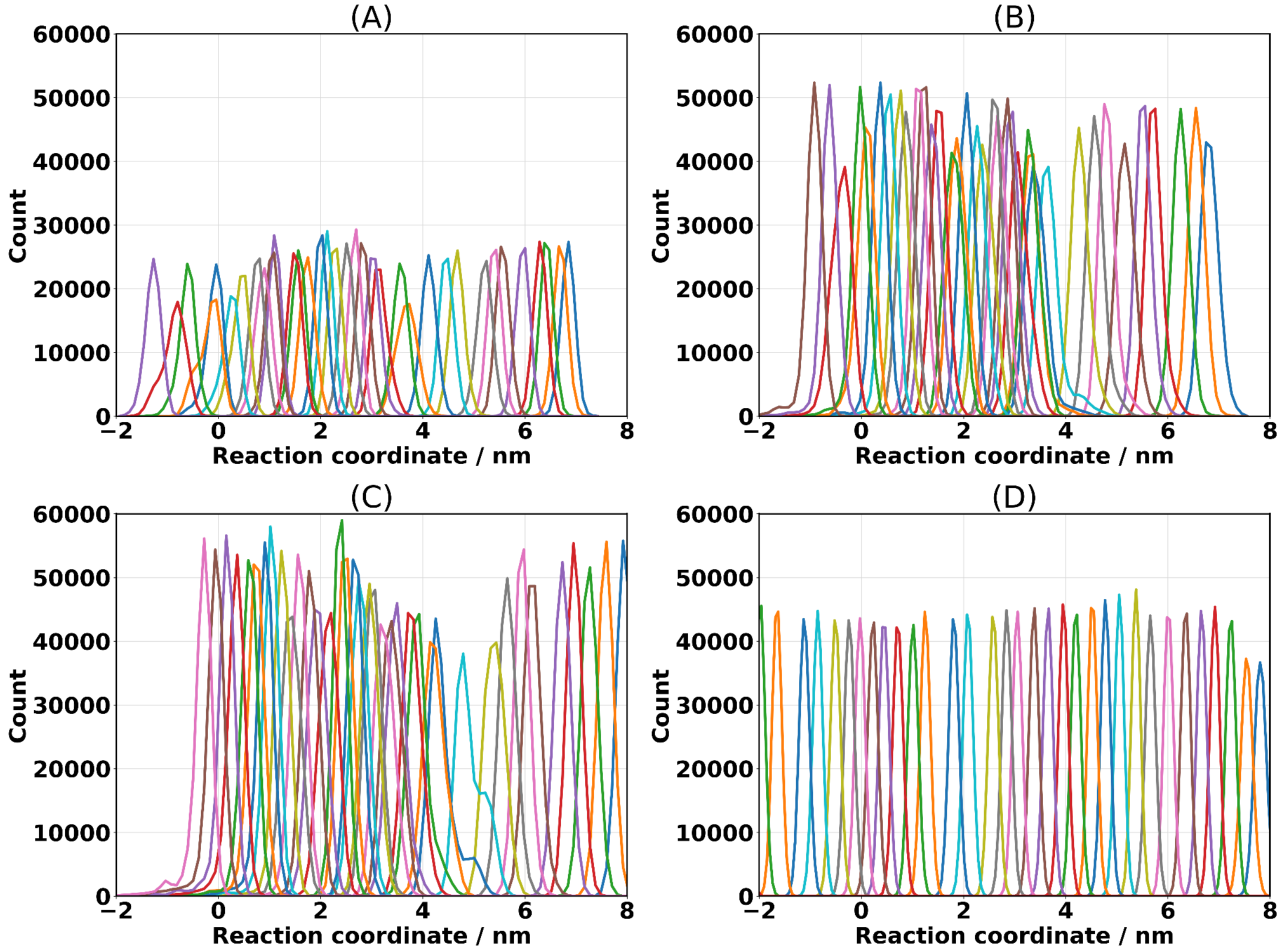

Figure S5 Configuration histograms corresponding to the simulation windows sampled with the umbrella sampling (US) method. These windows were collected from the reaction coordinate that describes the translocation processes of the polymers studied herein (pDMAEMA, pMETAC, pMEDSAH, and pSPMA) toward the center of the bacterial outer membrane (OM) model: pDMAEMA-OM (A), pMETAC-OM (B), pMEDSAH-OM (C), and pSPMA-OM (D). Count represents the number of configurations, whereas the reaction coordinate is the axis perpendicular to the plane of the OM.

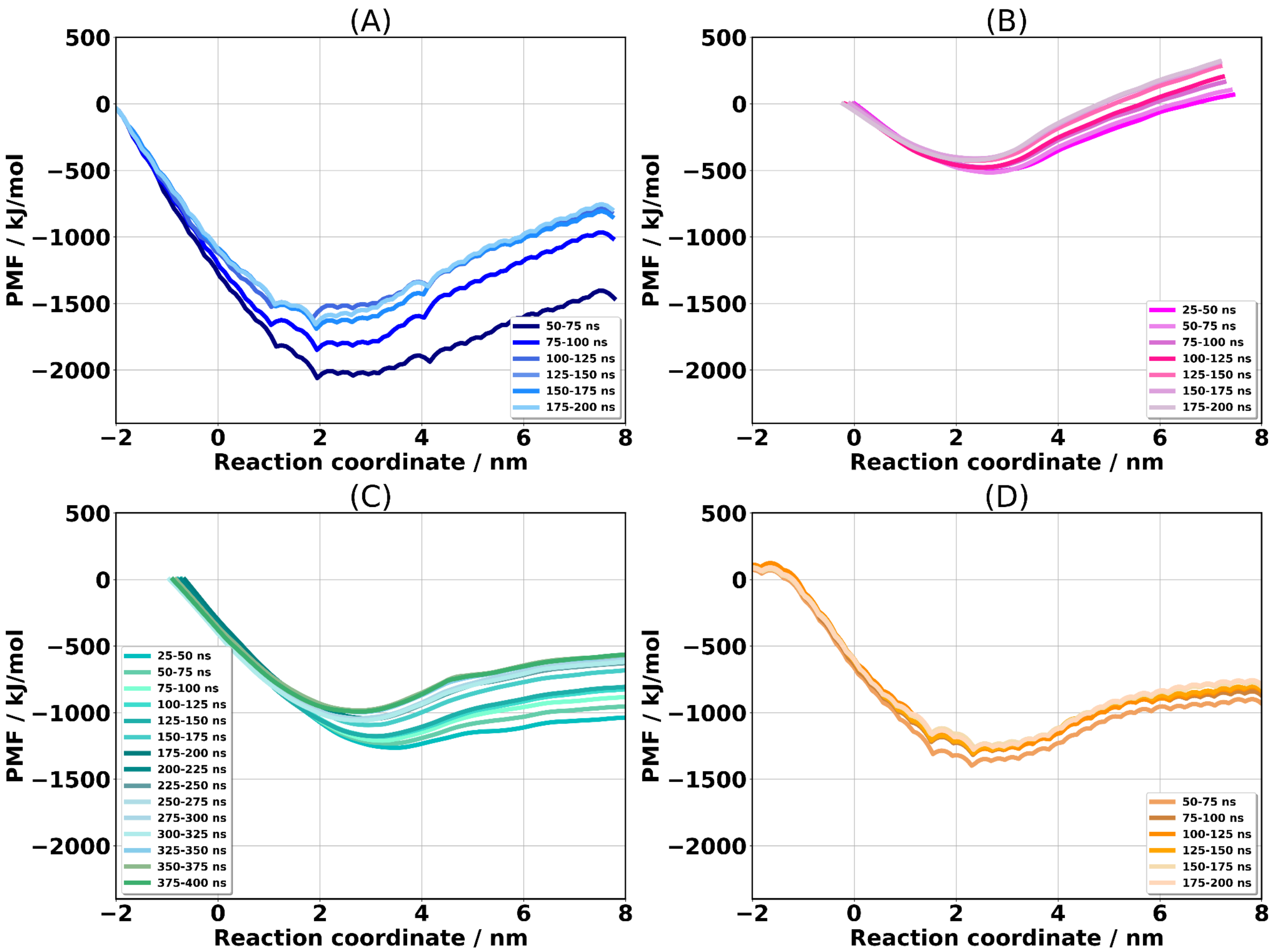

Figure S6 Block analysis of the potentials of mean force (PMFs) that describe the translocation processes of the polymers (pDMAEMA, pMETAC, pMEDSAH, and pSPMA) toward the bacterial outer membrane (OM) center.

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